Blackmon Lab

Evolutionary Biology

Relationship between chromosome number and genome size across Pteridophyta

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Background

- Chromosome number and genome size vary widely across plant lineages.
- Pteridophytes are known for having particularly high chromosome numbers compared to many other plant groups.
- However, genome size does not always scale proportionally with chromosome number.
- Genome size reflects total DNA content, while chromosome number reflects how that DNA is organized into chromosomes.
- Understanding the relationship between these two traits can provide insight into patterns of genome organization and evolution.

Research Question

Does genome size correlate with chromosome number across pteridophytes?

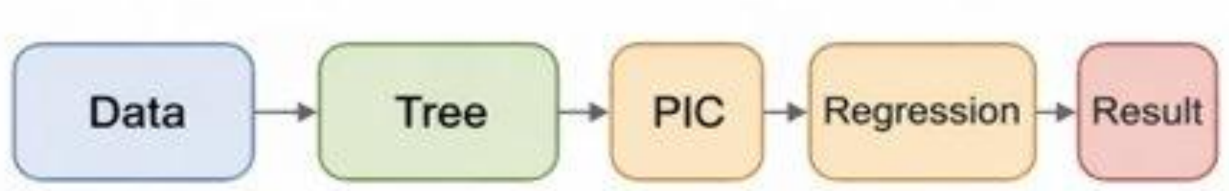
Study Design & AI

- Data collected for:
  genome size (1C, pg)
  haploid chromosome number (n)

- Species matched to a phylogenetic tree
- Data log-transformed

AI-assisted workflow:

- Supported debugging and structuring R code
- Assisted with phylogenetic analysis setup (tree + PIC)
- Results verified independently



Model Fitting

- Phylogenetically independent contrasts (PIC) were used to account for shared ancestry among species.
- PIC transforms species-level trait values into independent contrasts along branches of the phylogeny, allowing standard statistical methods to be applied without violating independence assumptions.
- Genome size (1C, pg) and haploid chromosome number (n) were log-transformed to improve linearity and reduce skew in the data.
- Species data were matched to a phylogenetic tree, and contrasts were calculated using this tree structure.
- Linear regression was performed on the contrasts:
  Chromosome number ~ genome size

Results

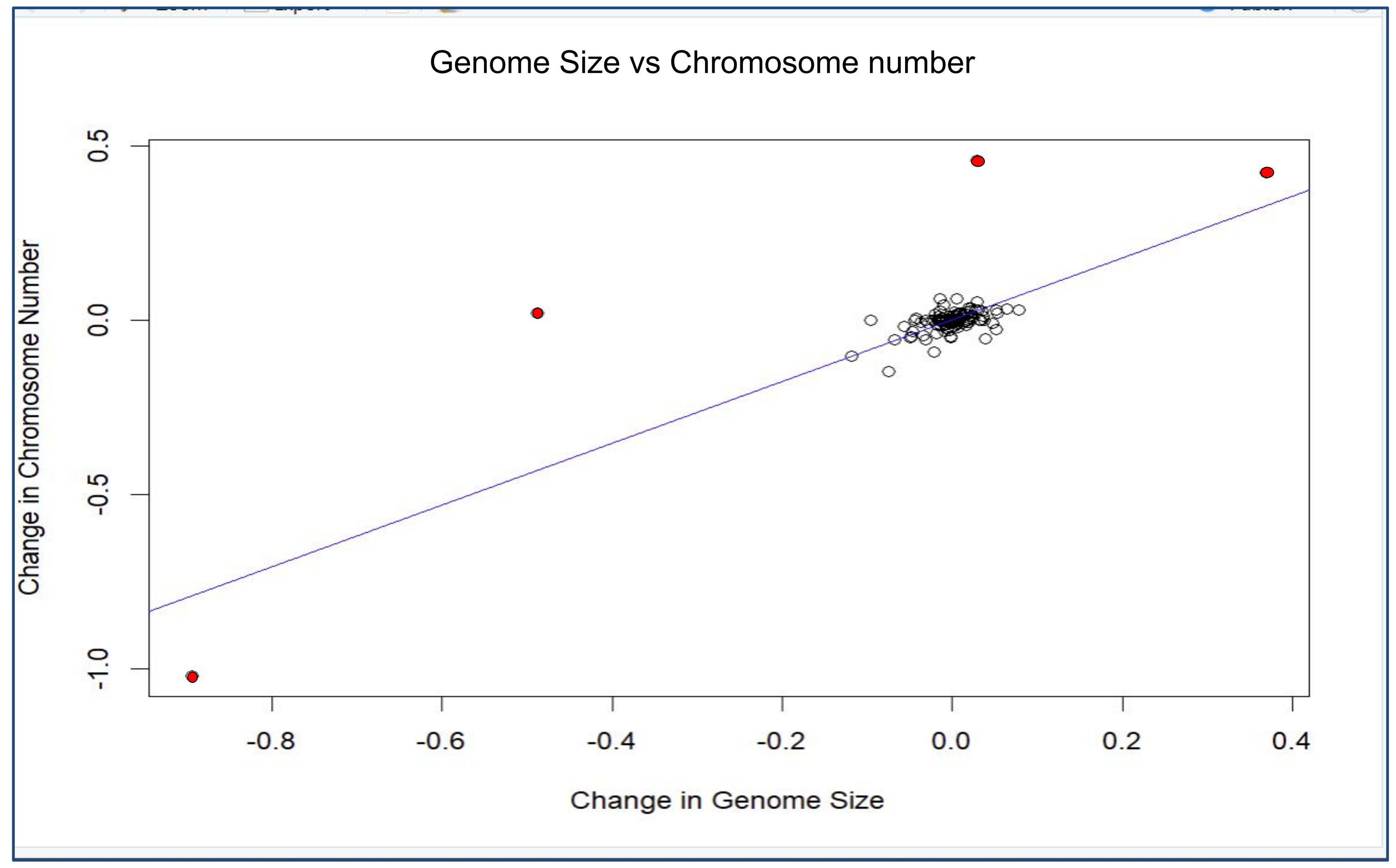


Figure 1. *Genome Size vs Chromosome Evolution (PIC Analysis)*

- n = 125 contrasts
- Slope = 0.885
- $R^2 = 0.65$
- $p < 2 \times 10^{-16}$

- Results show a strong positive relationship between genome size and chromosome number.
- Genome size explains approximately 65% of the variation in chromosome number.
- Species with larger genomes tend to have higher chromosome numbers.
- Results remained consistent after removing outliers, with a similar slope and increased R^2, indicating that the relationship is robust and not driven by extreme values.

Discussion

- Results show a strong positive relationship between genome size and chromosome number across pteridophytes.
- This indicates that species with larger genomes tend to have higher chromosome numbers.
- However, this analysis does not directly test the mechanisms responsible for this pattern.
- The relationship is not perfect ($R^2 = 0.65$), suggesting that additional factors contribute to variation in chromosome number.
- Limitations include the relatively small sample size due to limited overlap between genome size and chromosome datasets, as well as potential variability in publicly available data sources.
- Future work could include more species and analyses designed to directly test the mechanisms underlying this relationship.

Conclusions & Acknowledgements

- Genome size shows a strong positive relationship with chromosome number across pteridophytes.
- This indicates that species with larger genomes tend to have higher chromosome numbers.
- The relationship is robust and not driven by outliers.
- While this study does not directly test underlying mechanisms, one possible explanation is that increases in chromosome number are associated with increases in structural genomic elements, contributing to larger genome sizes.
- Overall, this study highlights a clear pattern between genome size and chromosome number and demonstrates the value of integrating phylogenetic methods with genomic data.

Thanks to Dr. Heath Blackmon and the Blackmon Lab for guidance, mentorship, and support throughout this project. Texas A&M University is acknowledged for providing the resources and research opportunities that made this work possible. This study utilized publicly available data from the Kew Plant DNA C-values Database, the Chromosome Counts Database (CCDB), and the Open Fern Tree of Life. AI tools (ChatGPT) were also used to assist with debugging R scripts, understanding phylogenetic methods, and refining data analysis workflows.